

MUTUAL FUND ANALYTICS PLATFORM

Final Report

Bluestock Fintech Internship | Data Analytics Track

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Tool Stack: Python | SQLite | Power BI | Jupyter Notebooks



1. Executive Summary

This report presents the Mutual Fund Analytics Platform developed for Bluestock Fintech. The platform analyses data from January 2022 to December 2025, covering 40 schemes across 10 fund houses and over 32,000 investor transactions from 5,000 unique investors.

Metric	Value
Total Schemes Analyzed	40 (34 Equity, 6 Debt)
Fund Houses Covered	10
Historical NAV Records	46,000 data points
Investor Transactions	32,778 (across 17 months)
Unique Investors	5,000
Industry AUM (Dec 2025 est.)	~₹70 Lakh Crore
Monthly SIP Inflow (Dec 2025)	₹31,002 Crore
Industry Folio Count (Dec 2025)	26.12 Crore

Key Highlights

- Monthly SIP inflows grew 170% over 4 years — from ₹11,438 Cr (Feb 2022) to ₹31,002 Cr (Dec 2025).
- Total industry folios nearly doubled from 13.26 Crore to 26.12 Crore, reflecting broad retail participation.
- Small Cap funds led on returns (~21.7% 3-year) while Liquid funds offered the best risk-adjusted performance (Sharpe: 5–8).
- Investors aged 26–35 dominated, accounting for ~41% of all transactions.
- T30 cities contributed 66% of all transaction volume.

2. Data Overview

2.1 Data Sources

The platform is built on 10 datasets sourced from AMFI data, MFAPI.in live NAV feeds, and internal investor transaction records, totalling ~4.6 MB and 127,650+ data points.

#	Dataset	Records	Key Fields
01	fund_master.csv	40 schemes	AMFI code, fund house, category, expense ratio, risk grade
02	nav_history.csv	46,000 rows	AMFI code, date, NAV
03	aum_by_fund_house.csv	90 rows	Fund house, AUM (₹Cr), scheme count
04	monthly_sip_inflows.csv	48 months	SIP inflow, active & new accounts
05	category_inflows.csv	144 rows	Month, category, net inflow (₹Cr)
06	industry_folio_count.csv	21 quarters	Total, equity, debt, hybrid folios
07	scheme_performance.csv	40 schemes	1Y/3Y/5Y returns, Sharpe, alpha, beta, drawdown
08	investor_transactions.csv	32,778 rows	Investor ID, type, amount, state, city tier, age, gender
09	portfolio_holdings.csv	322 rows	Stock symbol, sector, weight %, market value
10	benchmark_indices.csv	8,050 rows	NIFTY50, NIFTY100, BSE_SMALLCAP, CRISIL_GILT, etc.

2.2 Fund Universe

Category	Count	Avg AUM (₹Cr)	Avg Expense Ratio
Large Cap	7	23,176	~1.0%
Mid Cap	7	28,426	~1.2%
Small Cap	5	32,872	~1.3%
Flexi Cap	2	26,462	~1.0%
ELSS	4	2,989	~1.0%
Liquid	3	35,245	0.2%
Index / ETF	5	13,817	~0.2%
Gilt	2	27,066	~0.5%

2.3 Investor Demographic Profile

Dimension	Key Finding
Gender	66.5% Male 33.5% Female
Top Age Group	26–35 yrs (41.1%), followed by 36–45 (24.8%)
City Tier	T30 cities — 66% of transactions
Transaction Types	SIP 60.2% Lumpsum 24.7% Redemption 15.2%
Payment Modes	Banking 25.2% Cheque 25.1% UPI 24.9% Mandate 24.9%

3. ETL Pipeline

3.1 Architecture

The pipeline has three stages: Data Ingestion → Data Cleaning & Transformation → Load to SQLite DB.

- Ingestion: Live NAV data fetched via MFAPI.in REST API (live_nav_fetch.py) for 5 large-cap benchmark schemes.
- Cleaning: Handled in notebooks 01 and 02 — date standardization, currency normalization, categorical encoding, derived columns (YoY SIP growth), outlier checks on NAV values.
- Loading: Cleaned data loaded into bluestock_mf.db via SQLAlchemy (load_to_db.py) using pandas to_sql with if_exists='replace'.

3.2 Database Schema

Table	Type	Rows	Description
dim_fund	Dimension	40	Fund master — AMFI code, fund house, scheme name, category
fact_nav	Fact	46,000	Daily NAV history per scheme
investor_transactions	Fact	32,778	Full investor transaction ledger
scheme_performance	Fact	40	Aggregated performance metrics

3.3 Key SQL Queries

Ten analytical queries were developed: top 5 schemes by AUM, average NAV trend, SIP inflow aggregation, state-wise transaction volume, low-cost schemes (expense ratio < 1%), top 10 by Sharpe Ratio, category average annual return, underperforming schemes, NAV history for ICICI Pru Bluechip, and industry redemption volume.

4. EDA Findings

4.1 SIP Industry Growth

Period	Monthly SIP Inflow
Jan 2022	₹11,517 Cr
Dec 2022	₹13,573 Cr
Dec 2023	₹17,610 Cr
Dec 2024	₹26,459 Cr
Dec 2025	₹31,002 Cr

SIP monthly inflows grew 170% in 4 years, reflecting a significant structural shift from fixed deposits to equity investing among Indian retail investors.

4.2 AUM by Fund House

Fund House	AUM (₹Cr)
SBI Mutual Fund	12,50,000
ICICI Prudential MF	10,74,000
HDFC Mutual Fund	9,30,000
Nippon India MF	7,00,000
Kotak Mahindra MF	5,80,000

The top 3 fund houses control ~60% of total AUM, indicating moderate industry concentration.

4.3 Sector Allocation in Equity Funds

Sector	Aggregate Weight
Banking	652.26
IT	455.47
Pharma	407.45
Automobile	323.65

4.4 SIP Continuation Risk

Using a >45-day contribution gap as the at-risk threshold: 1,332 investors flagged as at risk vs. 30 active. The average SIP gap is 64.9 days — meaning investors are effectively missing every alternate SIP installment. This points to a significant retention problem requiring proactive engagement.

4.5 Category Net Inflows

Category	Net Inflow (₹Cr)
Flexi Cap	4,947
Large & Mid Cap	4,214
Mid Cap	3,897

Small Cap	3,533
Large Cap	2,413

5. Performance Analysis

5.1 Returns Summary

Metric	Mean Value
1-Year Return	14.38%
3-Year Return	14.09%
Sharpe Ratio	~1.1
Alpha	~1.2
Max Drawdown	-19.20%

5.2 Top Performers by 3-Year Return

Scheme	3-Year Return	Max Drawdown
SBI Small Cap Fund	23.39%	-30.87%
ABSL Small Cap Fund	22.38%	-29.15%
Axis Small Cap Fund	20.98%	-27.19%
Nippon India Small Cap	20.15%	-25.65%

All top performers are Small Cap funds delivering 8–11 percentage points above the Large Cap average — at the cost of significantly higher drawdown risk.

5.3 Best Risk-Adjusted Performers (Sharpe Ratio)

Scheme	Sharpe Ratio
ICICI Pru Liquid Fund	7.68
Kotak Liquid Fund	6.18
ABSL Liquid Fund	5.14
HDFC Short Term Debt Fund	1.84
SBI Magnum Gilt Fund	1.52

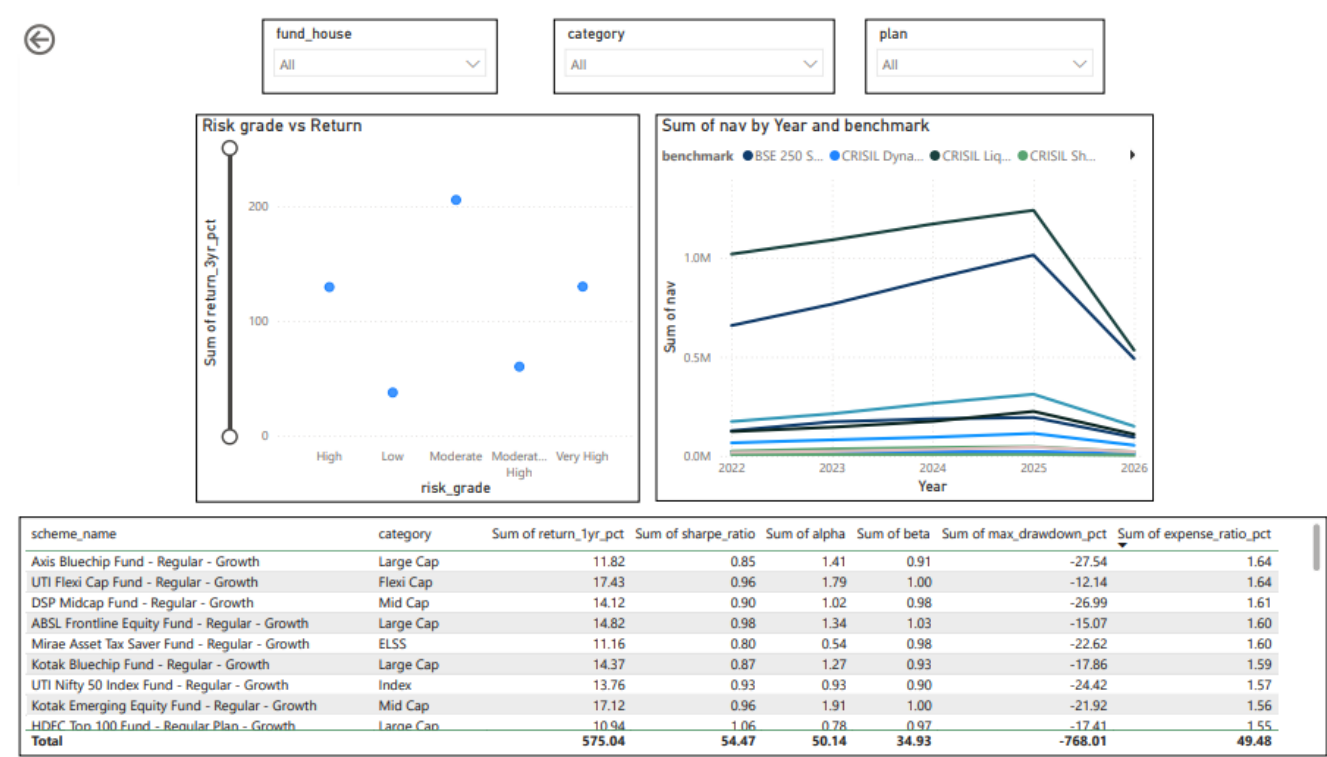
5.4 Category-Wise Risk vs. Return

Category	3-Year Return
Small Cap	21.69%
Mid Cap	16.59%
Flexi Cap	15.50%
Value	14.76%
Large & Mid Cap	14.56%

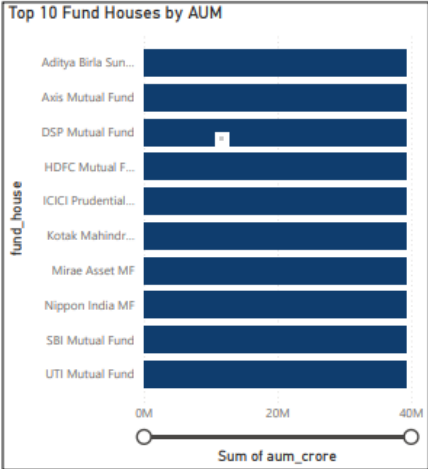
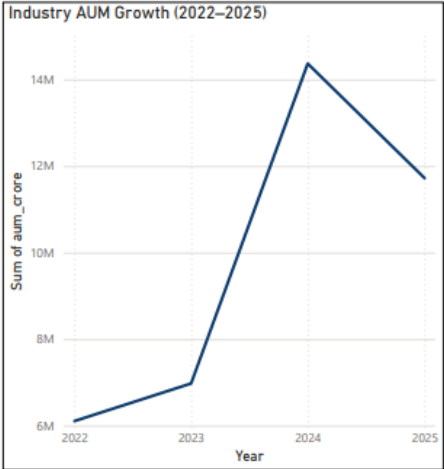
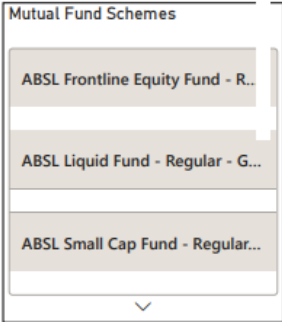
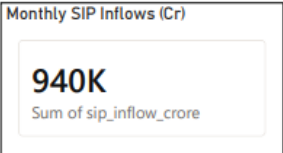
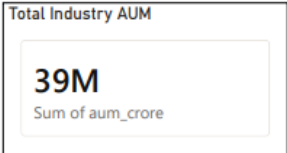
6. Dashboard Overview

The Power BI dashboard (bluestock_mf.pbix) contains four pages:

Page	Title	Key Visuals
1	Industry Overview	SIP inflow trend, folio count growth, AUM by fund house, category-wise net inflow
2	Fund Performance Scorecard	Top 10 by 3-year return, Sharpe vs return scatter, alpha leaderboard, expense ratio vs return frontier
3	Investor Behavior Analytics	Transaction type split, state heatmap, age & gender distribution, payment mode breakdown, T30 vs B30 split
4	Risk & Advanced Analytics	VaR/CVaR comparison, HHI concentration heatmap, max drawdown vs beta scatter, SIP continuity gauge, sector allocation sunburst



Bluestock Mutual Fund Industry Dashboard



state

All

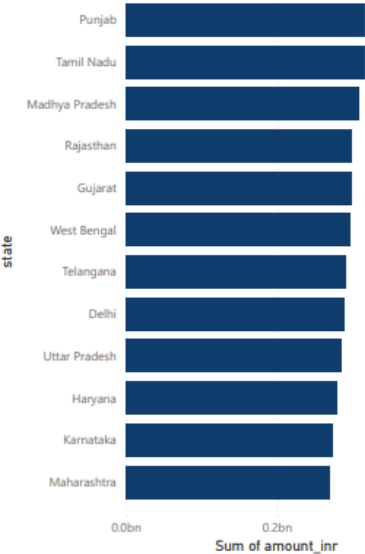
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All

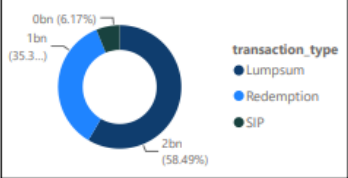
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All

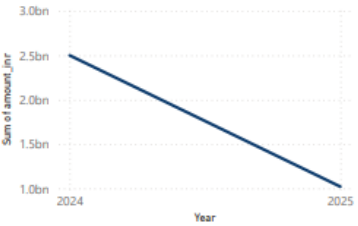
Transaction Amount by State



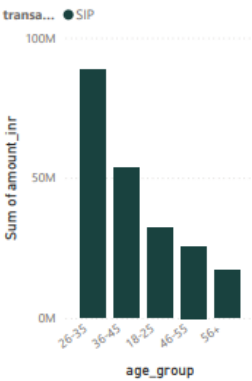
Transaction Type Split

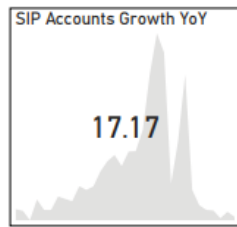


Monthly Transaction Volume



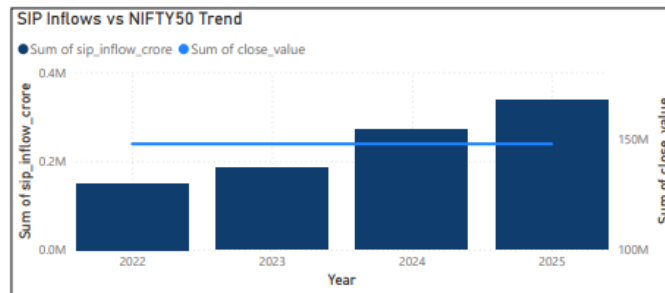
Sum of amount_inr by age_group and transaction_type



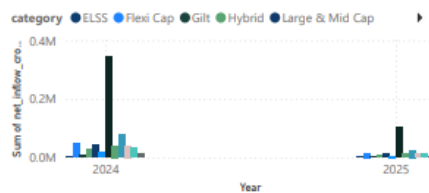


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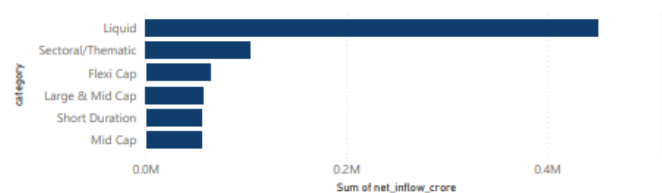
SIP Growth YoY



Category-wise Monthly Inflows



Top 5 Categories by Net Inflow FY25



7. Recommendations

For Fund Distributors & AMCs

- Target B30 cities with digital SIP campaigns to capture the next wave of retail investors.
- Implement SIP retention programs — average gap of 65 days indicates ~98% of SIP investors are at-risk of lapsing.
- Expand outreach to female investors (currently only 33.5% of the base) via targeted campaigns.

For Investors

- Allocate Flexi Cap funds as the core portfolio holding for balanced risk-return exposure.
- Use Liquid funds (Sharpe 5–8) as an emergency corpus or short-term parking vehicle.
- Small Cap allocation is suitable only for investors with a 5+ year horizon given drawdowns of 25–31%.

Platform Enhancements

- Implement real-time NAV alerting for threshold-based notifications.
- Build a rule-based fund recommender using risk profile and goal mapping.
- Automate ETL refresh to keep data current with daily AMFI feeds.
- Complete the compute_metrics.py and recommender.py scripts (currently placeholders).

8. Limitations

Limitation	Impact
Data is synthetic	Findings are directionally valid but not production-grade
Scheme universe limited to 40 funds	Does not represent the full ~1,500+ scheme market
AUM data at 9 point-in-time snapshots	Trend analysis is approximated rather than continuous
NAV history starts Jan 2022	Long-term pre-2022 cycle performance not captured
Portfolio holdings are a single snapshot	Sector allocations may not reflect current positions
VaR uses historical-parametric method	Tail risk during extreme events may be underestimated

Appendix: Tools & Libraries

Tool / Library	Version	Purpose
Python	3.13	Core scripting and data processing
pandas	Latest	Data manipulation and transformation
SQLAlchemy	Latest	Database ORM and connection management
SQLite	3.x	Lightweight local database engine
Power BI Desktop	Latest	Dashboard creation and visualization
Jupyter Notebook	Latest	Exploratory data analysis
requests	Latest	HTTP calls to MFAPI.in REST API
MFAPI.in	Public	Live NAV data source

Project File Structure: Mutual-Fund-Analytics-Platform/ contains data/ (raw, processed, db), scripts/ (live_nav_fetch.py, load_to_db.py, etl_pipeline.py, compute_metrics.py, recommender.py), notebook/ (5 Jupyter notebooks), sql/ (schema.sql, queries.sql), dashboard/ (.pbix and .pdf), and reports/ (this report and Presentation.pptx).